Herarchical connection with MCU

U_Power
Power.SchDoc

USB_DP — USB2_P
USB_DN — USB2_N
VBAT_MON — VBAT_MON
RXD0 — RXD0
TXD0 — TXD0

U_Pyros
Pyros.SchDoc

PYRO_CH1_CONT — PYRO_CH1_CONT
PYRO_CH2_CONT — PYRO_CH2_CONT
PYRO_CH1_GATE — PYRO_CH1_GATE
PYRO_CH2_GATE — PYRO_CH2_GATE

U_RF_LoRa_&_GPS
RF_LoRa_&_GPS.SchDoc

SNS_SPI_MOSI — SNS_SPI_MOSI
SNS_SPI_MISO — SNS_SPI_MISO
SNS_SPI_SCK — SNS_SPI_SCK
RFM95W_CS — RFM95W_CS
RFM95W_RESET — RFM95W_RESET
RFM95W_INT — RFM95W_INT
GPS_UART_TX — GPS_UART_TX
GPS_UART_RX — GPS_UART_RX
PPS_GPS — PPS_GPS
INT_GPS — INT_GPS
RESET_GPS — RESET_GPS

U_Sensors
Sensors.SchDoc

MPU6050_INT — MPU6050_INT
SNS_I2C_SCL — SNS_I2C_SCL
SNS_I2C_SDA — SNS_I2C_SDA
SDO — SDO

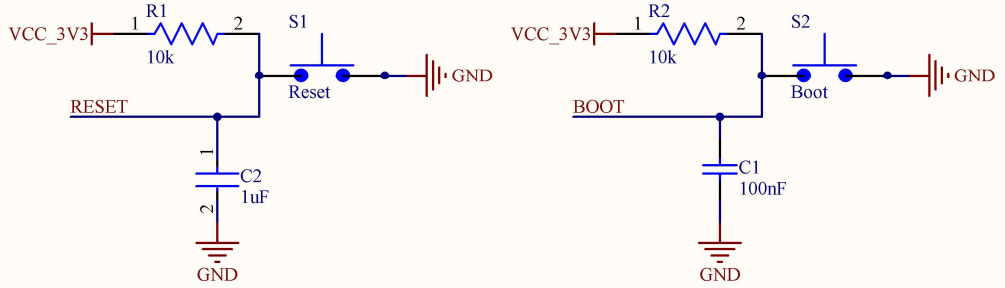
U_Storage
Storage.SchDoc

SD_DETECT — SD_DETECT
SD_CS — SD_CS
SNS_SPI_SCK — SNS_SPI_SCK
SNS_SPI_MISO — SNS_SPI_MISO
SNS_SPI_MOSI — SNS_SPI_MOSI

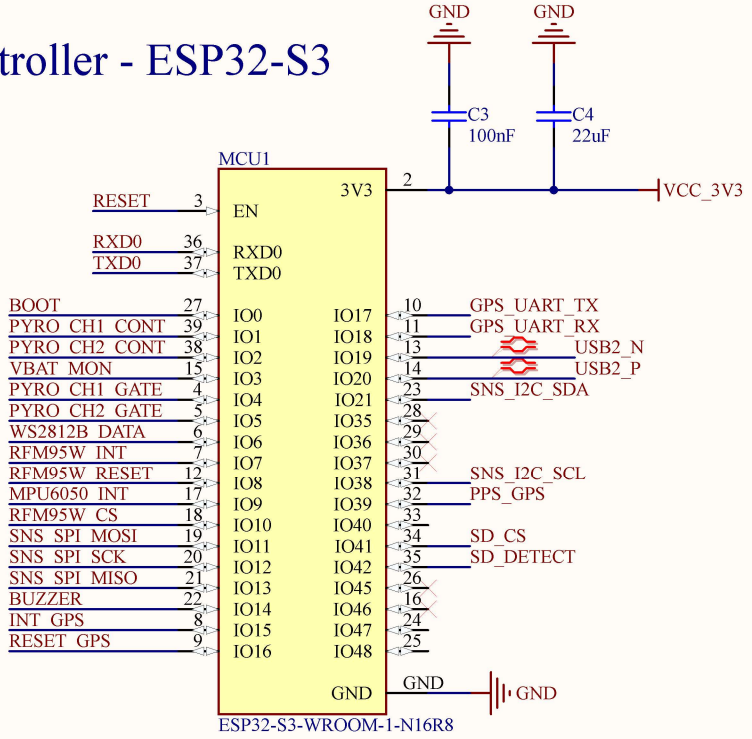
U_Indicators
Indicators.SchDoc

BUZZER — BUZZER
WS2812B_DATA — WS2812B_DATA

Reset & Boot Buttons

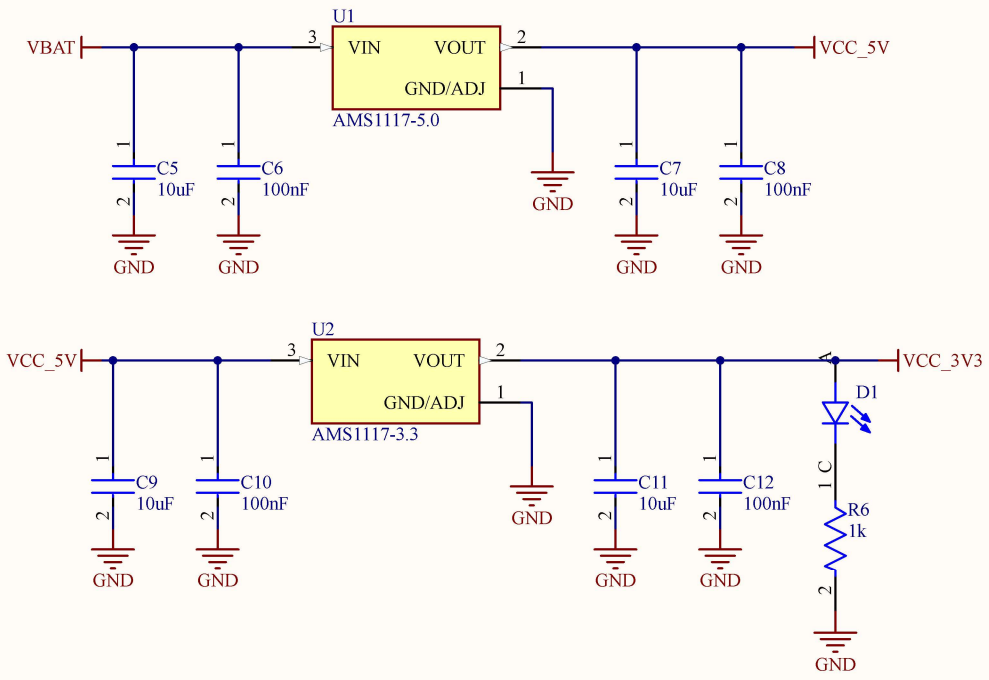


Microcontroller - ESP32-S3

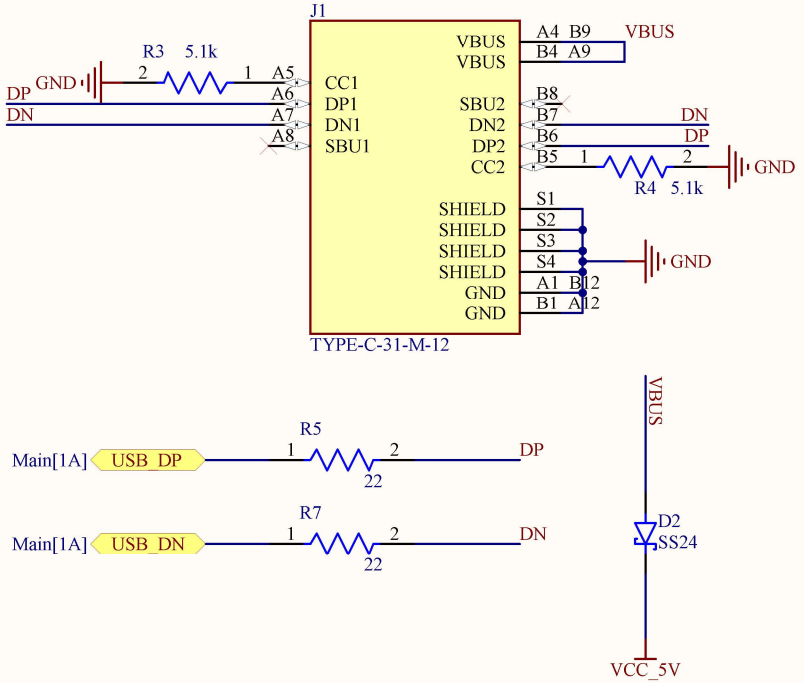


Title Main — ESP32-S3 Microcontroller & System Interconnect		IGNITIA — Tecnológico de Monterrey
Size A	Number IGNIS-v5-SCH-01	Revision v5.0
Date: 11/10/2026		Sheet 1 of 7
File: C:\Users\..\Main.SchDoc		Drawn By: Emilio Guadarrama

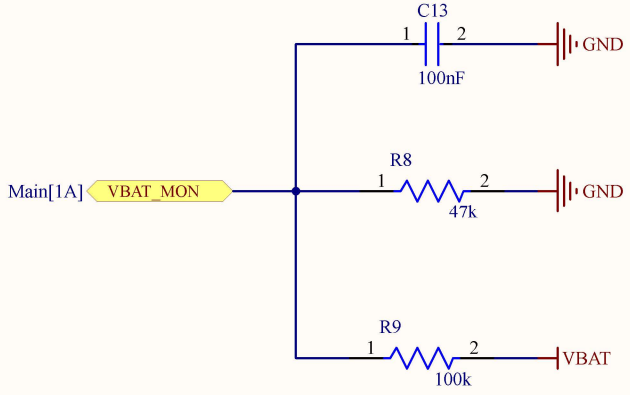
Dual Stage Voltage Regulation



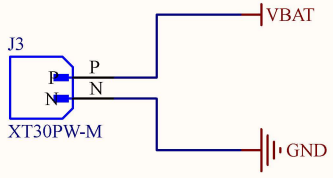
USB Type C Port - Data/Power



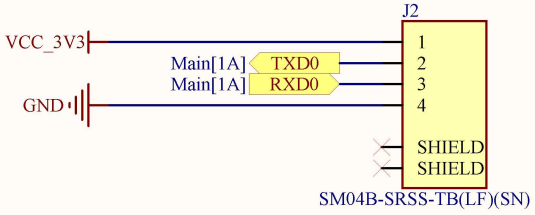
Voltage Monitoring



Battery Connector

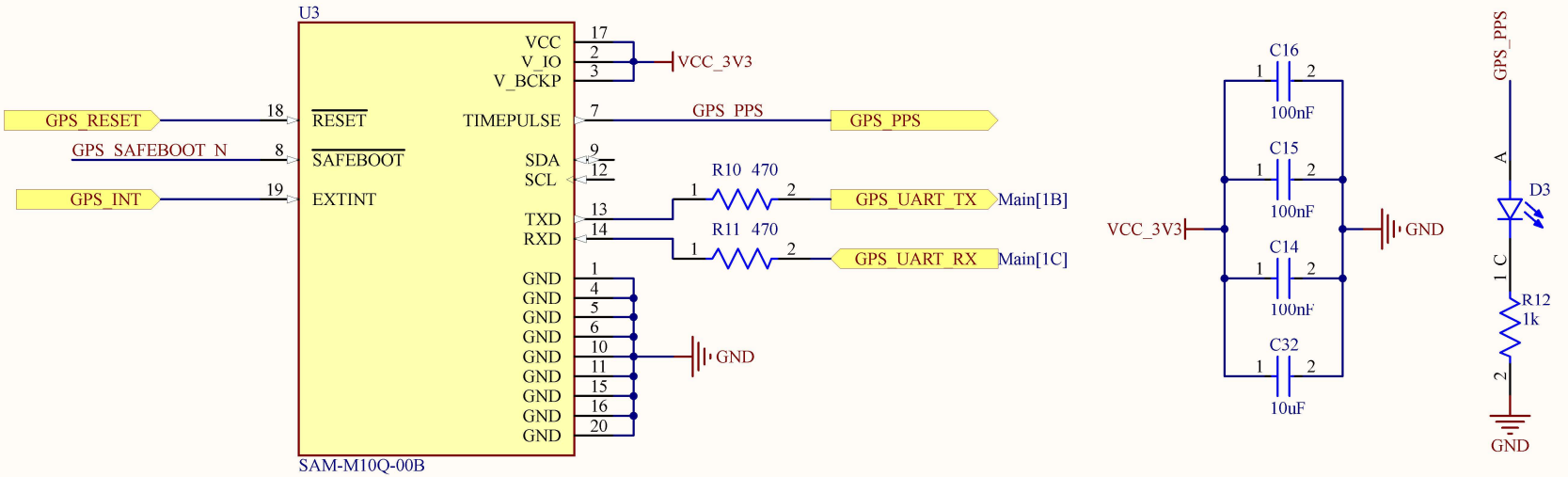


JST Connector - UART

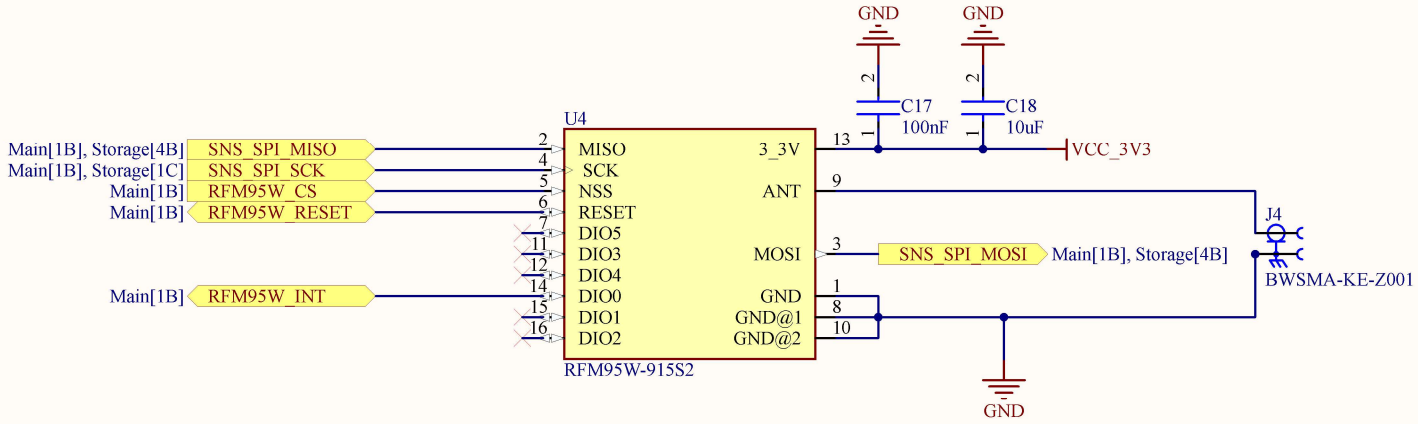


Title Power Supply — Dual Stage Regulation & USB-C		IGNITIA
Size A	Number IGNIS-v5-SCH-02	Revision v5.0
Date: 11/10/2026		Sheet 2 of 7
File: C:\Users\..\Power.SchDoc		Drawn By: Emilio Guadarrama

GPS Module — SAM-M10Q-00B

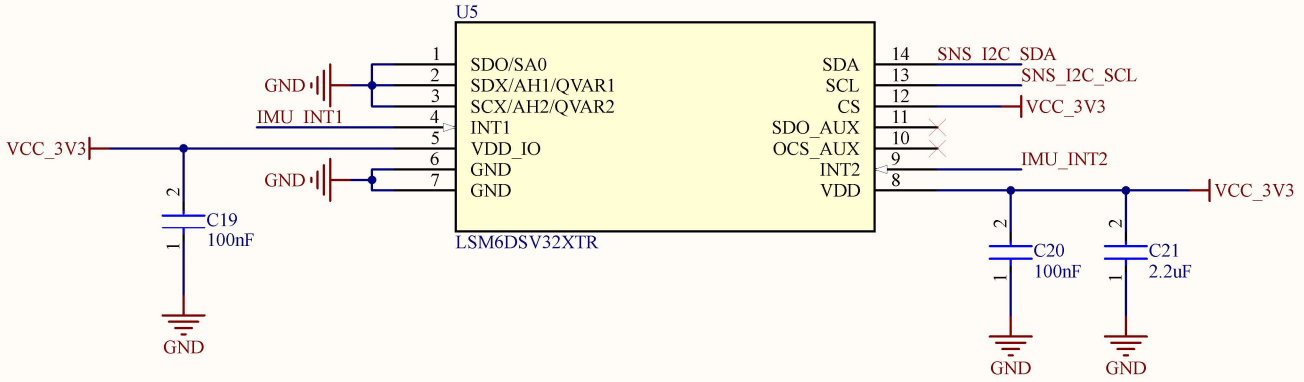


LoRa Transceiver — RFM95W 915 MHz

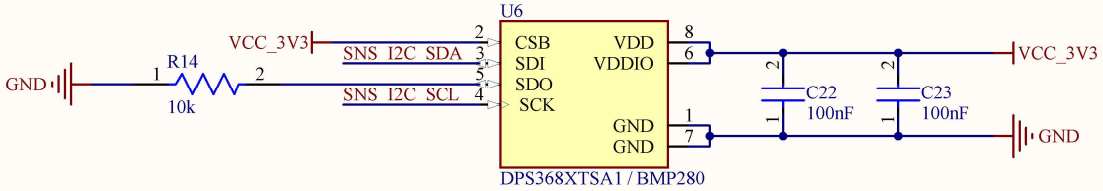


Title RF Telemetry System & GPS		
Size A	Number IGNIS-v5-SCH-03	Revision v5.0
Date: 11/10/2026		Sheet 3 of 7
File: C:\Users\..\RF_LoRa_&_GPS.SchDoc		Drawn By: Emilio Guadarrama

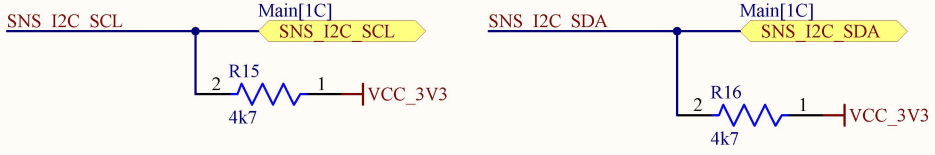
Inertial Measurement Unit — ST LSM6DSV32XTR



Barometric Altimeter — DPS368XTSA1 / BMP280

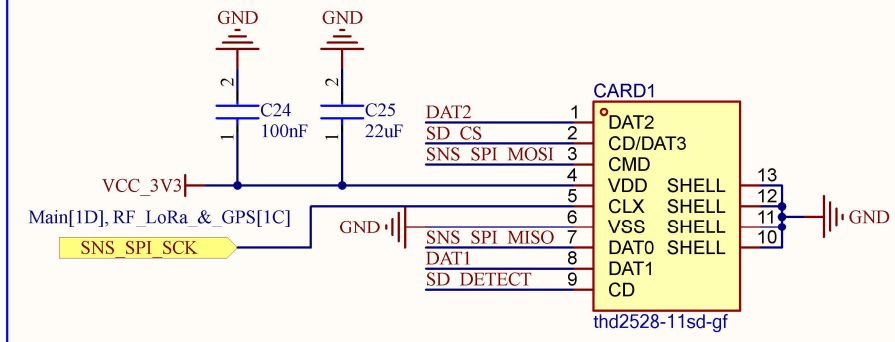


I2C Bus — Signal Routing

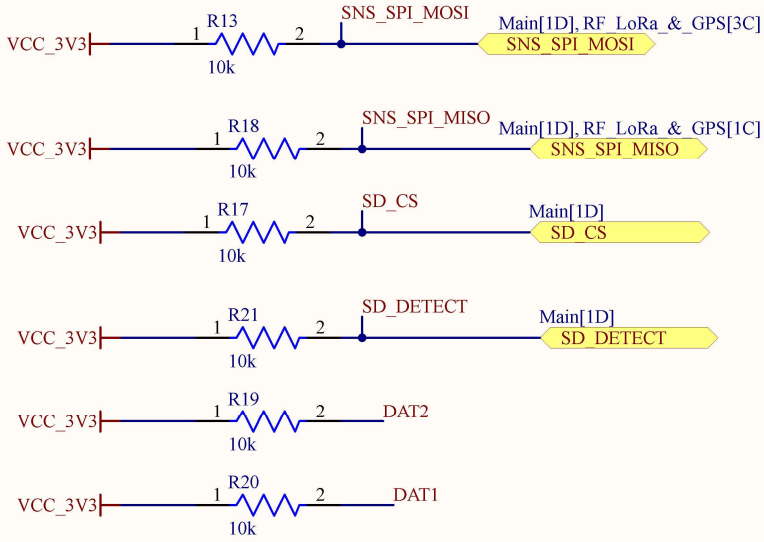


Title	Environmental & Inertial Sensors	IGNITIA / Tecnológico de Monterrey	
Size A	Number IGNIS-v5-SCH-04		Revision v5.0
Date:	11/10/2026		Sheet 4 of 7
File:	C:\Users\..\Sensors.SchDoc		Drawn By: Emilio Guadarrama

MicroSD Card Slot — THD2528-11SD-GF

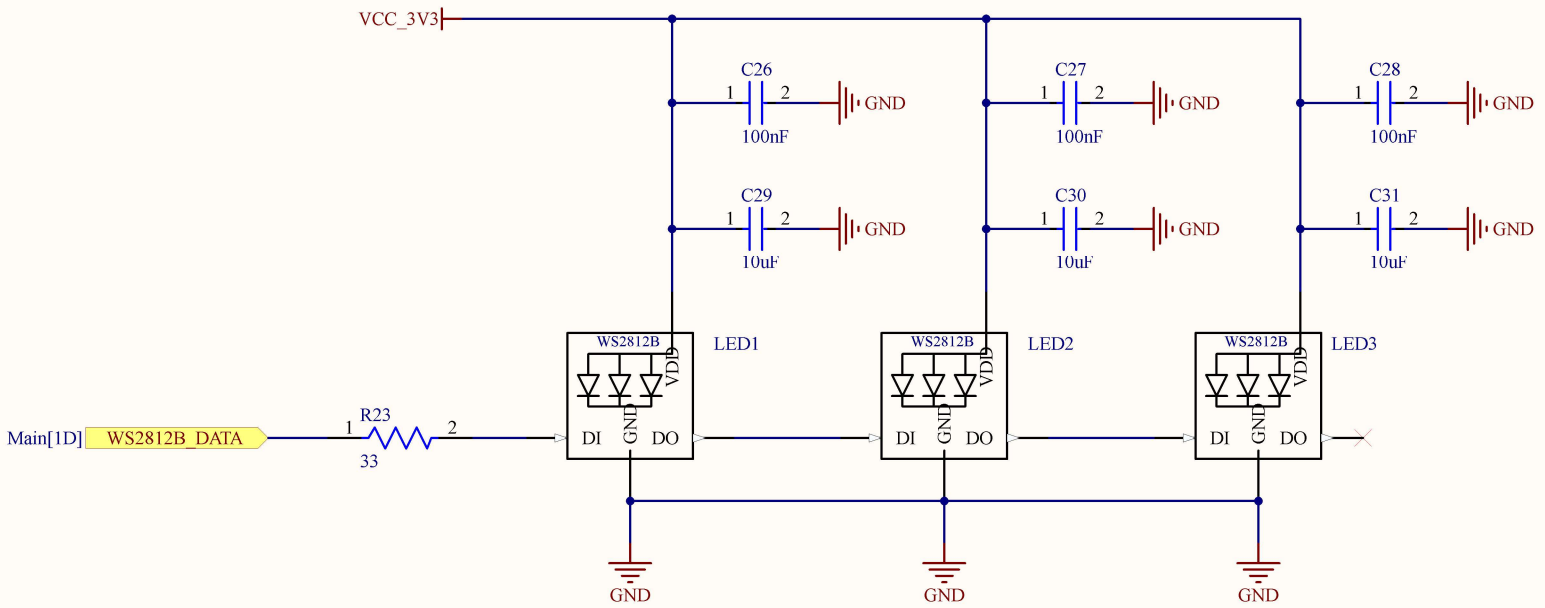


Pull UP Resistances

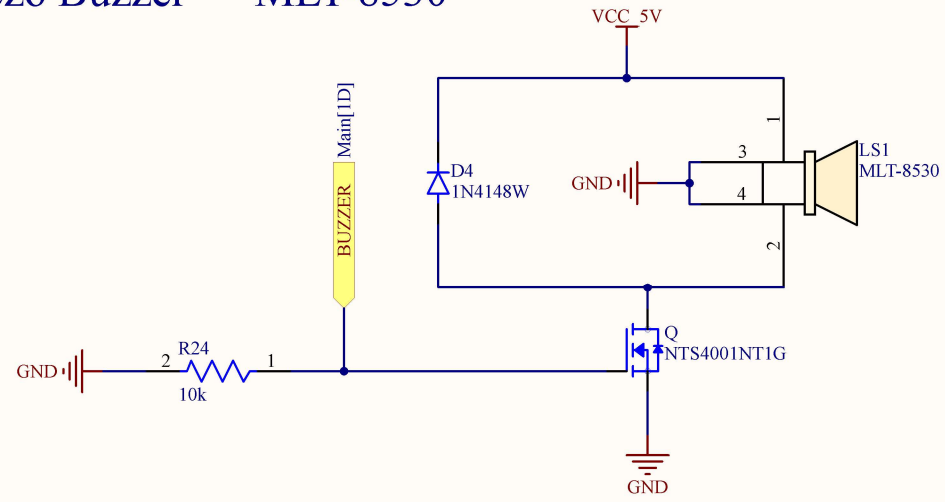


Title Mass Storage — MicroSD		IGNITIA / Tecnológico de Monterrey
Size A	Number IGNIS-v5-SCH-05	Revision v5.0
Date: 11/10/2026		Sheet 5 of 7
File: C:\Users\..\Storage.SchDoc		Drawn By: Emilio Guadarrama

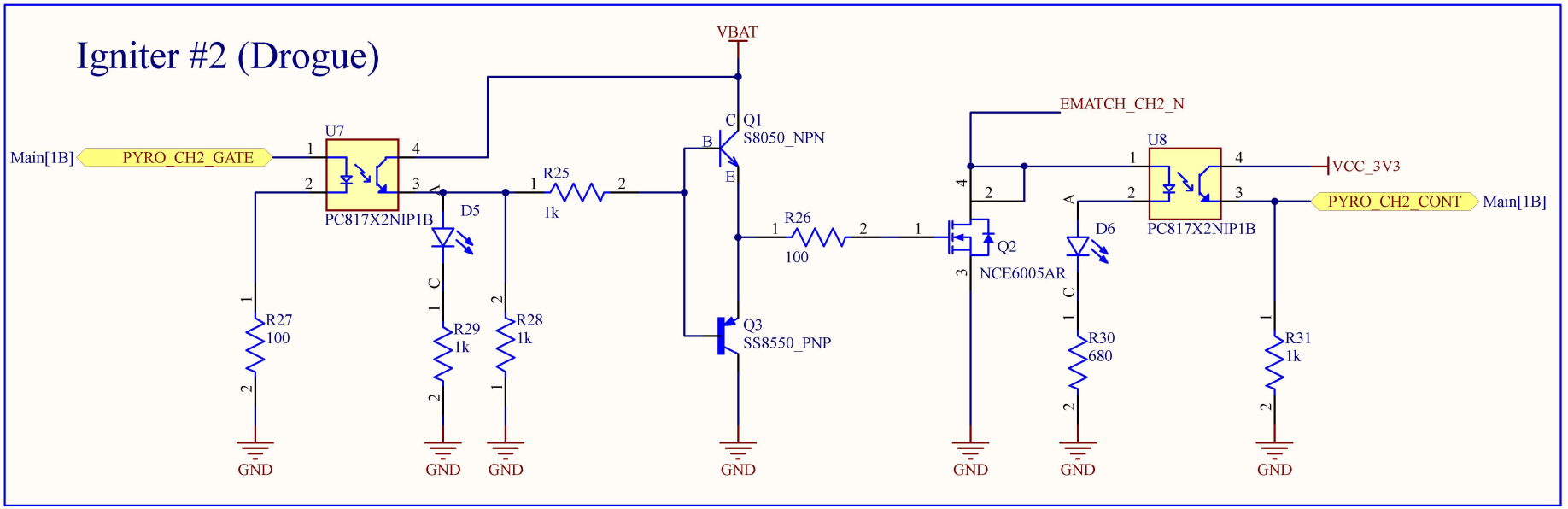
RGB Status Indicators — WS2812B

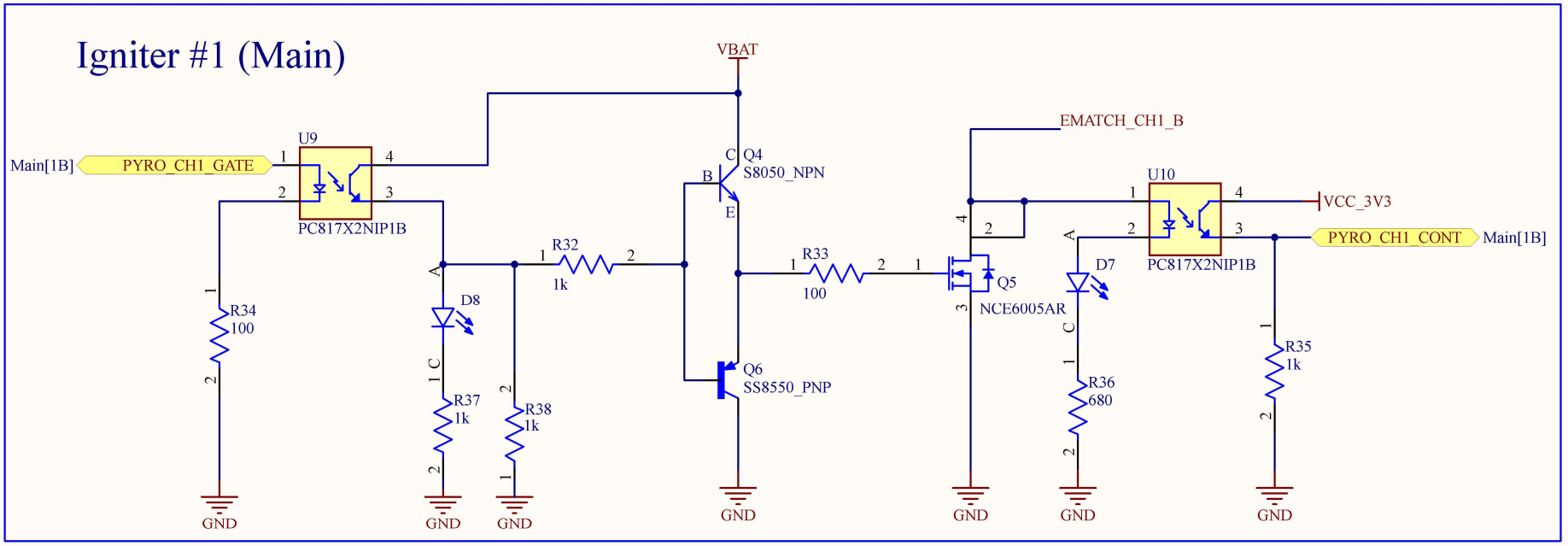


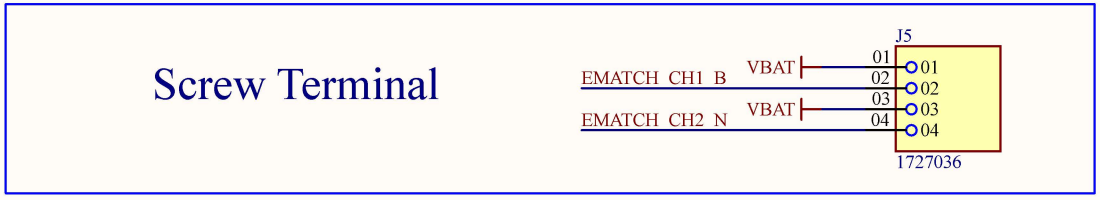
Piezo Buzzer — MLT-8530



Title Status Indicators		IGNITIA / Tecnológico de Monterrey	
Size A	Number IGNIS-v5-SCH-06		Revision v5.0
Date: 11/10/2026		Sheet 6 of 7	
File: C:\Users\..\Indicators.SchDoc		Drawn By: Emilio Guadarrama	







Title Pyrotechnic Deployment System		IGNITIA / Tecnológico de Monterrey
Size A	Number IGNIS-v5-SCH-07	Revision v5.0
Date: 11/10/2026		Sheet 7 of 7
File: C:\Users\..\Pyros.SchDoc		Drawn By: Emilio Guadarrama